

Charles Lepine

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EDUCATION

North Carolina State University | Raleigh, NC

B.S. Computer Engineering | Senior Standing | Aug 2022 - Dec 2026

Relevant Coursework: Data Structures & OOP (C++), Computer Systems Programming, Introduction to Computer Systems, Computer Networking, Discrete Mathematics, Fundamentals of Logic Design, Introduction to Embedded Systems

In Progress (Fall 2026): Microprocessor Architecture, Design of Complex Digital Systems, Communication Engineering, Senior Design

TECHNICAL SKILLS

Languages: C, C++, Java, Python, PowerShell

Frameworks & Libraries: LangChain, LiteLLM, Pydantic, Posh-ACME

Tools & Platforms: Git, GitHub, VS Code, Wireshark, IBM HTTP Server, GSKit, Ollama

Cloud & Infrastructure: AWS, Azure, Google Cloud, Cloudflare (DNS/ACME integrations), TLS/PKI

EXPERIENCE

IBM | Software Developer Intern - RTP, NC

May 2026 - Aug 2026

SSL/TLS Certificate Lifecycle Automation

- Built a PowerShell automation system for IBM HTTP Server that cut TLS certificate deployment from a 1-3 day manual process to a 5-minute setup, with every subsequent renewal then completing automatically and unattended in a few minutes.
- Designed a modular challenge-plugin architecture of 6 plugins supporting HTTP and DNS validation across AWS, Azure, Cloudflare, and Google Cloud, making the tool compatible with any ACME-compliant certificate authority on the market.
- Consolidated 20 manual renewal steps into a single script with encrypted credential resolution from cloud vaults, automatic rollback on failed deployment, and email alerting on every renewal event; validated across 13 separate production domains.

AI Agent Extraction Pipeline

- Designed a three-stage pipeline (Slack ingestion, preprocessing, and LLM extraction) that applies PII and secrets guardrails and turns raw engineering discussion threads into a structured, searchable FAQ page in under 10 minutes of runtime.
- Orchestrated three sequential LLM calls (question extraction, answer extraction, and topic classification) in LangChain via LiteLLM, enforcing consistent, machine-parsable structured output with a Pydantic schema at every stage of the chain.
- Extended the pipeline to additional source types with an adaptive segmentation algorithm that reused the LLM extraction stage unmodified, and added a checkpoint system so that repeat runs skip any threads that were already processed.

NCFC / Wake FC / Net Sports | Soccer Referee - Raleigh-Durham, NC

June 2020 - Present

- Officiated over 500 matches from amateur leagues through semi-professional UPSL play, enforcing FIFA and league laws.
- Mentored junior and assistant referees on positioning, mechanics, and rule interpretation to ensure consistency.

PROJECTS

AI Occupancy Detection Camera - NVIDIA PeopleNet, DeepStream SDK

Senior Design | Aug 2026 - Present

- Leading a team of 4 building an AI occupancy camera that counts people in a room and streams live headcount to a dashboard.
- Architecting the vision pipeline around NVIDIA PeopleNet for person detection, deployed through the NVIDIA DeepStream SDK.
- Owning scope breakdown and integration across the team, from camera capture through inference to the dashboard front end.

Remote-Operated Embedded Vehicle - C, MSP430

Embedded Systems | Course Project

- Developed embedded firmware in C for an MSP430 microcontroller to drive and steer a remote-operated vehicle.
- Implemented modules for infrared sensing, motion control algorithms, LCD display output, and IoT communication.
- Interpreted manufacturer datasheets to integrate hardware peripherals and write low-level device drivers from scratch.

LEADERSHIP & ACTIVITIES

NC State Club Soccer - Team Member | Jan 2024 - Present

- Selected from over 200 candidates in a competitive 3-day tryout; competes in the CCS league and regional tournaments.

NC State Rocket League Esports - Team Member | Aug 2022 - Present

- Competed on NC State's highest-ranked Rocket League team, reaching a top 0.01% global ranking and winning tournaments.